

Homework 3 Solutions

Definitions

Define the following terms/concepts. (The language of probability - sample space, event, random trial, union, intersection, random variable, probability distribution and the Law of Large Numbers - was covered on **Homework 2**; review those definitions before starting.)

- a.) Population distribution - is the probability distribution for a single observation of a random variable
- b.) Sampling distribution - the probability distribution of a statistic (such as $\bar{x}$ or $\hat{p}$) computed from a sample of size n , describing how that statistic varies from sample to sample
- c.) Probability Mass Function - A function which assigns a probability to each outcome of a discrete random variable
- d.) Probability Density Function - A function which assigns probability density to different outcomes of a continuous random variable

Sampling Distributions, Means, and Variances

(1) A random variable X has the following probability distribution

X	P(X)
1	0.25
2	0.25
3	0.25
4	0.25

- Derive the sampling distribution for the mean of $n = 2$ observations of X

The sample space for a mean of two observations of X is given below.

Observation 1	Observation 2	$\bar{X}$
1	1	1.0
2	1	1.5
3	1	2.0
4	1	2.5
1	2	1.5
2	2	2.0
3	2	2.5
4	2	3.0
1	3	2.0
2	3	2.5
3	3	3.0
4	3	3.5
1	4	2.5
2	4	3.0
3	4	3.5
4	4	4.0

The sampling distribution of $\bar{X}$ is given by:

$\bar{X}$	$P(\bar{X})$
1.0	$1(0.25^2) = 0.0625$
1.5	$2(0.25^2) = 0.125$
2.0	$3(0.25^2) = 0.1875$
2.5	$4(0.25^2) = 0.25$
3.0	$3(0.25^2) = 0.1875$
3.5	$2(0.25^2) = 0.125$
4.0	$1(0.25^2) = 0.0625$

- Compute the mean $\mu_{\bar{X}}$ and standard deviation $\sigma_{\bar{X}}$ of the sampling distribution of $\bar{X}$

Using the sampling distribution of $\bar{X}$ derived above:

$$\mu_{\bar{X}} = \sum_{\bar{x}} \bar{x}P(\bar{x})$$

$$= 1(0.0625) + 1.5(0.125) + \cdots + 3.5(0.125) + 4(0.0625)$$

$$= 2.5$$

$$\sigma_{\bar{X}} = \sqrt{\sum_{\bar{x}} (\bar{x} - \mu_{\bar{X}})^2 P(\bar{x})}$$

$$= \sqrt{(1 - 2.5)^2(0.0625) + (1.5 - 2.5)^2(0.125) + \cdots + (3.5 - 2.5)^2(0.125) + (4 - 2.5)^2(0.0625)}$$

$$= \sqrt{0.625}$$

$$= 0.791$$

Note that this agrees with the population values for X itself: $\mu = 2.5$ and $\sigma = \sqrt{1.25} = 1.118$, and the standard deviation of the sampling distribution is $\sigma/\sqrt{n} = 1.118/\sqrt{2} = 0.791$.

(2) A random variable Y has two possible outcomes. The outcome $Y = 1$ occurs with probability $p = 0.23$. Derive the sampling distribution for the proportion of outcomes with $Y = 1$ in a sample of $n = 3$ observations of Y .

The population distribution of Y is given by:

Y	$P(Y)$
0	0.77
1	0.23

The sample space for the sample proportion of 3 observations of Y is given below.

Observation 1	Observation 2	Observation 3	$\hat{p}$
0	0	0	0.00
1	0	0	0.33
0	1	0	0.33
1	1	0	0.67
0	0	1	0.33
1	0	1	0.67
0	1	1	0.67
1	1	1	1.00

The sampling distribution of $\hat{p}$ is given by:

$\hat{p}$	$P(\hat{p})$
0.00	$1(0.77^3) = 0.457$
0.33	$3(0.23 \times 0.77^2) = 0.409$
0.67	$3(0.23^2 \times 0.77) = 0.122$
1.00	$1(0.23^3) = 0.012$

(3) Consider a survey of tree volume. Suppose that in the population of trees 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let X denote the volume of a randomly selected tree, assuming that every tree has an equal chance of being selected. The probability distribution of X is given in the table below.

X	$P(X)$
10	0.1
20	0.2
30	0.3
40	0.4

- On **Homework 2** you confirmed that X has a mean of $\mu = 30$ cubic feet and a standard deviation of $\sigma = 10$ cubic feet. Derive the sampling distribution for the mean tree volume in a sample of $n = 2$ trees.

The sample space for the sample mean of 2 observations of tree volume is given below.

Observation 1	Observation 2	$\bar{X}$
10	10	10
20	10	15
30	10	20
40	10	25
10	20	15
20	20	20
30	20	25
40	20	30
10	30	20
20	30	25
30	30	30
40	30	35
10	40	25
20	40	30
30	40	35
40	40	40

The sampling distribution of $\bar{X}$ is given by:

$\bar{X}$	$P(\bar{X})$
10	$(0.1)^2 = 0.01$
15	$2(0.1 \times 0.2) = 0.04$
20	$2(0.1 \times 0.3) + 0.2^2 = 0.1$
25	$2(0.4 \times 0.1) + 2(0.2 \times 0.3) = 0.2$
30	$2(0.2 \times 0.4) + 0.3^2 = 0.25$
35	$2(0.3 \times 0.4) = 0.24$
40	$(0.4)^2 = 0.16$

Deriving Probabilities From Discrete Distributions

(4) Michael Dickson, who plays for the Seattle Seahawks, is a punter in the NFL. The following sampling distribution gives the mean punting distance (in yards) for a sample of $n = 4$ punts. Use this distribution to answer the following questions:

$\bar{X}$	Mean Punt Distance	$P(\bar{X})$
35.33		0.40
45.67		0.50
55.00		0.08
60.33		0.01
65.67		0.01

- What is the probability the mean distance is 55 or more yards?

$$\begin{aligned}
 P(\bar{X} \geq 55) &= 1 - P(\bar{X} < 55) \\
 &= 1 - (0.4 + 0.5) = 0.1
 \end{aligned}$$

- What is the probability the mean distance is at least 35.33 yards?

$$P(\bar{X} \geq 35.33) = 1$$

- What is the probability the mean distance is between 35.33 and 45.67 yards?

$$P(35.33 \leq \bar{X} \leq 45.67) = P(X = 35.33) + P(X = 45.67) = 0.9$$

- What is the probability the mean distance is 35.33 yards or 55 yards?

Let A be the event $\bar{X} = 35.33$ and B be the event $\bar{X} = 55$. These outcomes are disjoint, so

$$P(A \cup B) = P(\bar{X} = 35.33) + P(\bar{X} = 55) = 0.4 + 0.08 = 0.48$$

(5) A random variable X follows a binomial distribution with probability of success $p = 0.43$ and number of trials $n = 8$. What is the probability of observing fewer than 3 successes, that is what is $P(X < 3)$?

Let $X \sim \text{Binomial}(p = 0.43, n = 8)$ then

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \sum_{k=0}^2 \frac{8!}{k!(8-k)!} \times 0.43^k \times (1 - 0.43)^{8-k} \approx 0.256$$

(6) Tom Brady is a famous quarterback who recently retired from playing in the National Football League. During his career, the probability that Brady would complete a pass in a given play was 64.3%. If a typical NFL team runs about 35 offensive pass plays, what is the probability that Brady would complete exactly 28 pass plays in a game?

Let $X \sim \text{Binomial}(p = 0.643, n = 35)$ then

$$P(X = 28) = \frac{35!}{28!(35-28)!} \times 0.643^{28} \times (1 - 0.643)^{35-28} \approx 0.021$$

(7) A random variable W follows a Poisson distribution with rate parameter $\lambda = 1.8$ events per hour. What is the probability of observing exactly four events in a given hour, that is what is $P(W = 4)$?

Let $W \sim \text{Poisson}(\lambda = 1.8)$

$$P(W = 4) = \frac{1.8^4 \cdot e^{-1.8}}{4!} \approx 0.072$$

(8) A factory experiences on average 3 equipment malfunctions per day, following a Poisson distribution with a rate parameter of 3. Calculate the probability that on a randomly selected day, the factory will observe 3 or more equipment malfunctions.

let X represent the number of equipment failures in a given day for the factory, then

$$X \sim \text{Poisson}(\lambda = 3)$$

$$P(X \geq 3) = 1 - P(X < 3)$$

where

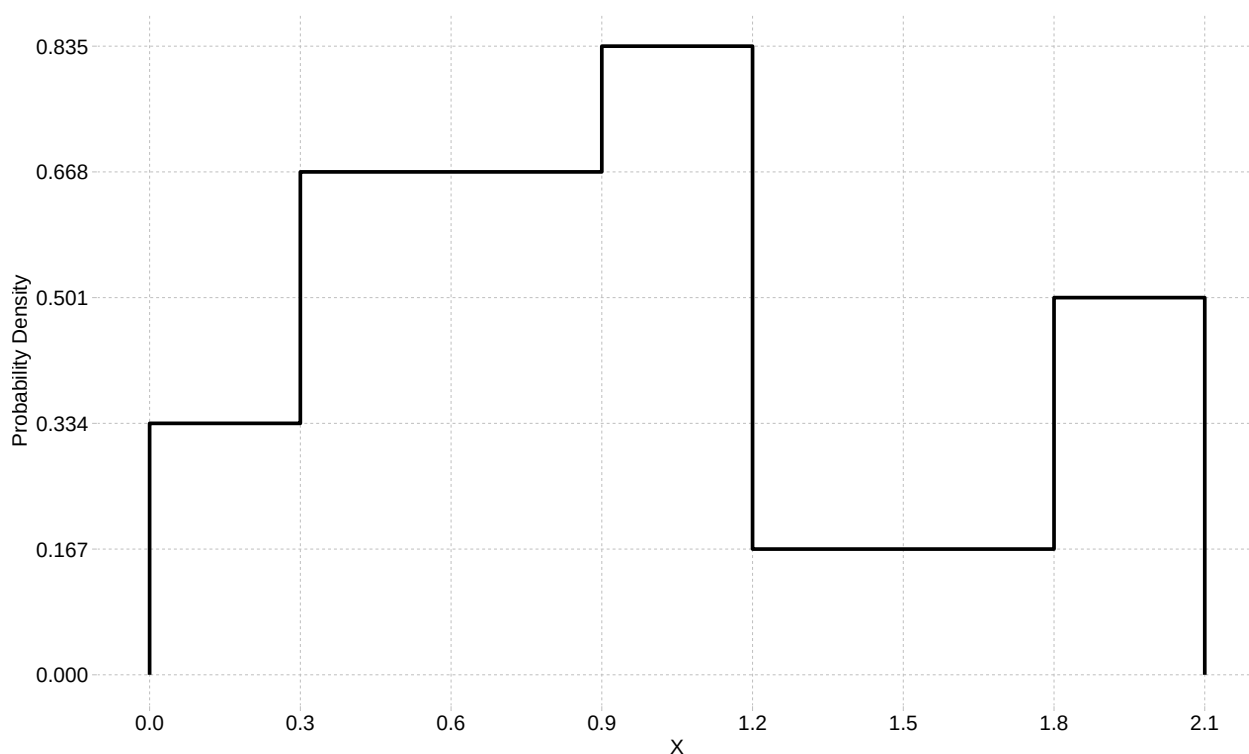
$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = \sum_{k=0}^2 \frac{3^k \cdot e^{-3}}{k!} \approx 0.423$$

so

$$P(X \geq 3) = 1 - 0.423 \approx 0.577$$

Deriving Probabilities From Continuous Distributions

(9) Use the plot below of the probability distribution of a continuous random variable X to answer the following questions



Use the plot to find the following probabilities:

- What is $P(X < 0.6)$?

In this plot the area comprising a single grid square is

$$\text{Area of a rectangle} = \text{Base} \times \text{Height} = 0.167 \times 0.3 \approx 0.05$$

there are six grid squares between 0 and 0.6, therefore

$$P(X < 0.6) = 6 \times 0.05 = 0.3$$

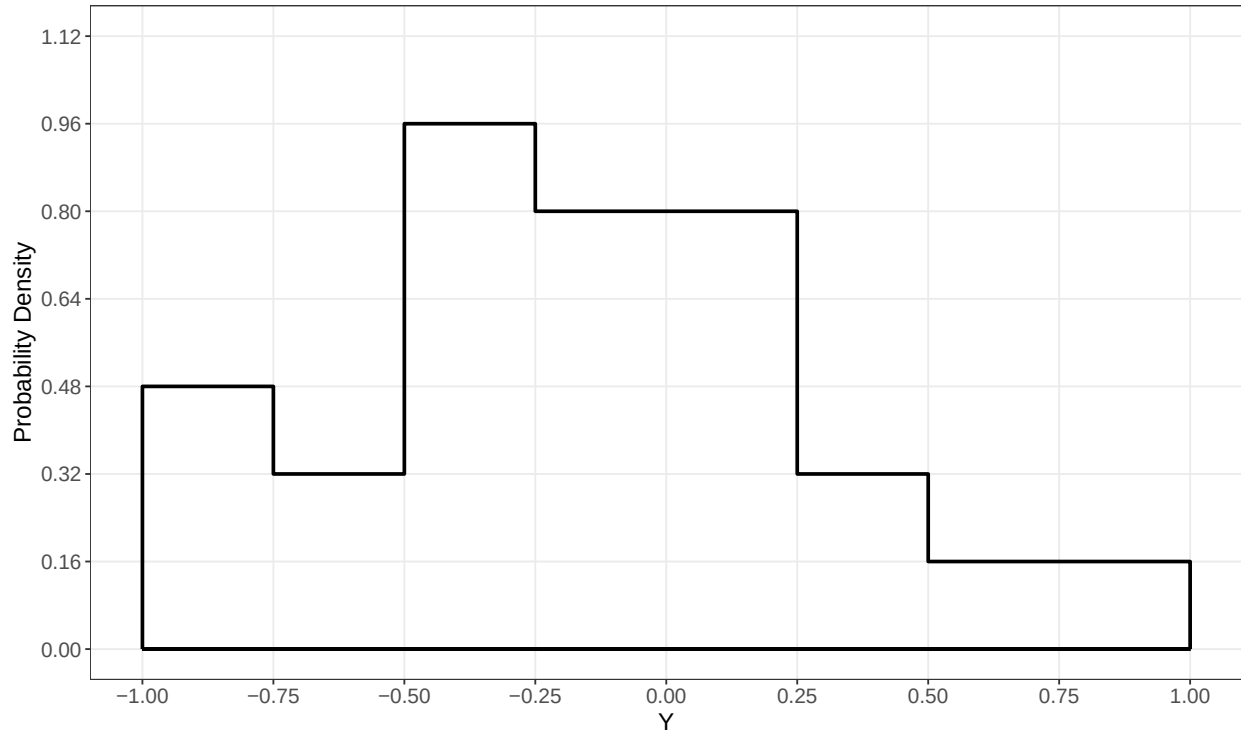
- What is $P(X > 0.3)$?

$$P(X > 0.3) = 1 - P(X \leq 0.3) = 1 - (2 \times 0.05) = 0.9$$

- What is $P(1.5 < X < 1.8)$?

$$P(1.5 < X < 1.8) = 0.05$$

(10) Use the plot below of the probability distribution of a continuous random variable Y to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$)

- What is $P(Y > -0.25)$?

In this plot the area comprising a single grid square is

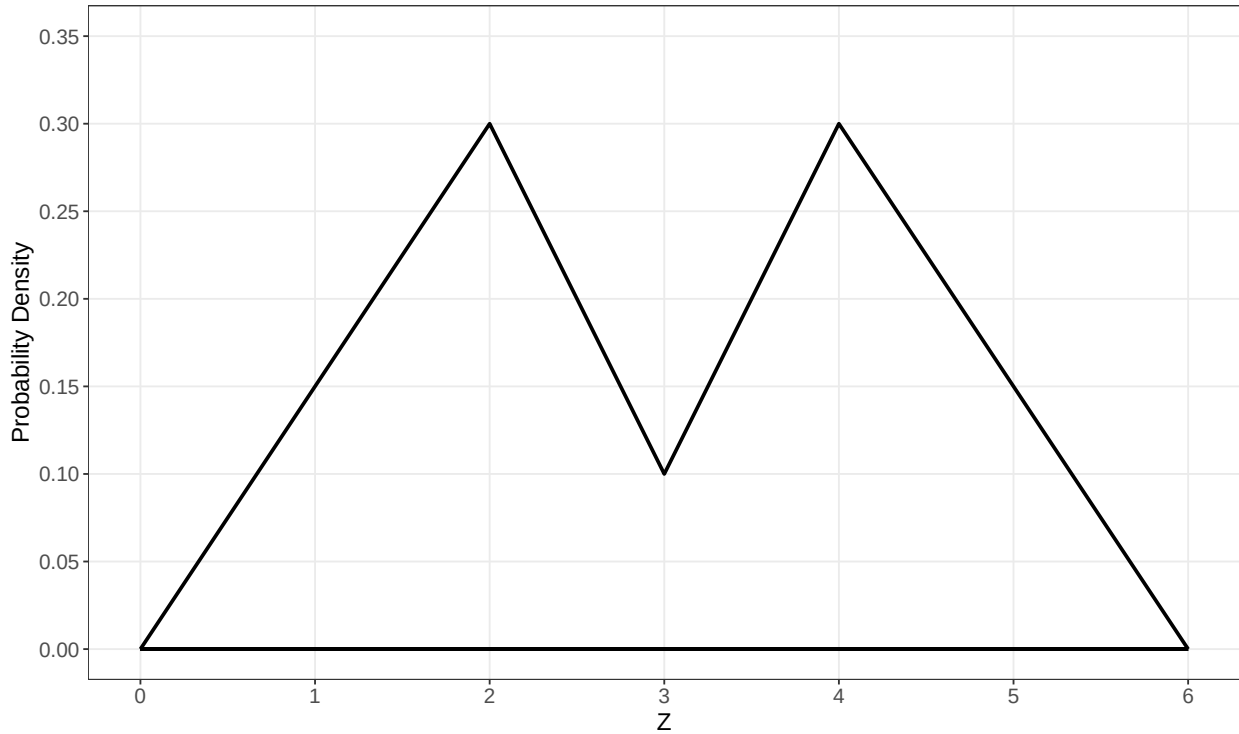
$$\text{Area of a rectangle} = \text{Base} \times \text{Height} = 0.25 \times 0.16 \approx 0.04$$

There are 14 grid squares in the area between -0.25 and 1.0 $P(Y > -0.25) = 14 \times 0.04 = 0.56$

- What is $P(-0.5 < Y < 0.25)$?

There are 16 grid squares in the area between -0.5 and 0.25 therefore $P(-0.5 < Y < 0.25) = 16 \times 0.04 = 0.64$

(11) Use the plot below of the probability distribution of a continuous random variable Z to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$)

- What is $P(Z < 2)$?

$$P(Z < 2) = 0.5(2)(0.3) = 0.3$$

- What is $P(2 < Z < 3)$?

In this plot a single grid square has an area of 0.05 thus the area between $Z = 2$ and $Z = 3$ is given by $P(2 < Z < 3) = 2(0.05) + (0.5)(1)(0.2) = 0.2$

- What is $P(Z > 4)$?

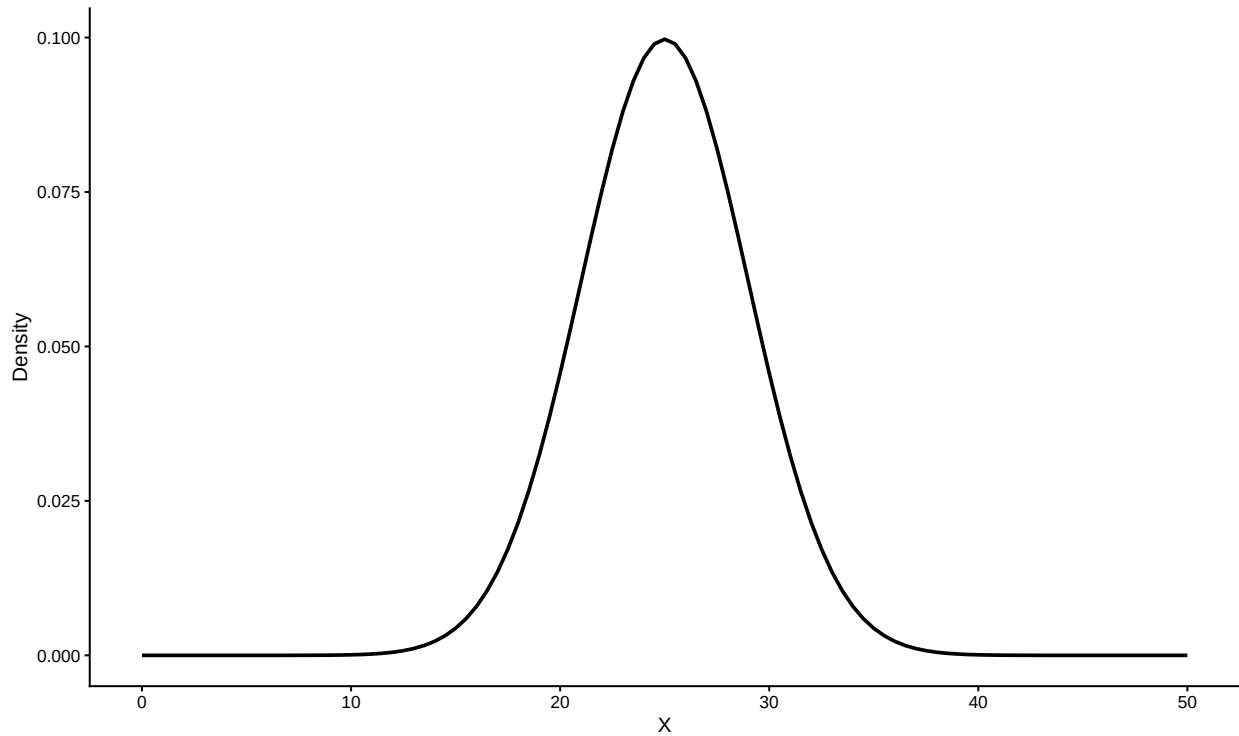
$$P(Z > 4) = P(Z < 2) = 0.3$$

- Give the approximate value of the 25th percentile of distribution of Z

Reading from the plot, the area to the left of $Z = 2$ is 0.3, so the 25th percentile lies just below 2. Working it out exactly: on $[0, 2]$ the density is the line $f(z) = 0.15z$, so the area to the left of z is $0.075z^2$. Setting $0.075z^2 = 0.25$ gives

$$Q1 = \sqrt{0.25/0.075} \approx 1.83$$

12.) Assume the continuous random variable X follows a normal distribution with mean $\mu = 25$ and variance $\sigma^2 = 16$.



Using a Z-table, find the following (approximate) probabilities for X . You can also use the app in the course website or StatDistributions.com to check your result

- What is $P(X < 15)$?

$$z = \frac{15 - 25}{4} = -2.5$$

$$P(Z < -2.5) = 0.006$$

- What is $P(X > 28)$?

$$z = \frac{28 - 25}{4} = 0.75$$

$$P(Z > 0.75) = 1 - P(Z \leq 0.75) = 1 - 0.773 = 0.227$$

- What is $P(12 < X < 20)$?

$$z_{\text{lower}} = \frac{12 - 25}{4} = -3.25$$

$$z_{\text{upper}} = \frac{20 - 25}{4} = -1.25$$

$$P(-3.25 < Z < -1.25) = P(Z < -1.25) - P(Z < -3.25) = 0.106 - 0.001 = 0.105$$

- What is $P(X > 30)$?

$$z = \frac{30 - 25}{4} = 1.25$$

$$P(Z > 1.25) = 1 - P(Z \leq 1.25) = 1 - 0.894 = 0.106$$

- What is $P(25 < X < 33)$?

$$z_{\text{lower}} = \frac{25 - 25}{4} = 0$$

$$z_{\text{upper}} = \frac{33 - 25}{4} = 2$$

$$P(0 < Z < 2) = P(Z < 2) - P(Z < 0) = 0.977 - 0.5 = 0.477$$

- What is the value of X that marks the 95th percentile of the distribution?

the 95th percentile of a standard normal distribution is $z = 1.645$ plugging this into the z -score equation and solving for X gives

$$(z \times \sigma) + \mu = (1.645 \times 4) + 25 = 31.58$$

- What is the value of X that marks the 40th percentile of the distribution?

the 40th percentile of a standard normal distribution is $z = -0.253$

$$(z \times \sigma) + \mu = (-0.253 \times 4) + 25 = 23.99 \approx 24.0$$

13.) Suppose that a random variable X follows a normal distribution with mean $\mu = -5$ and standard deviation $\sigma = 1.8$. Using the plot of the Z -distribution below, shade in the area corresponding to the probability $P(-2 < X < 0.5)$

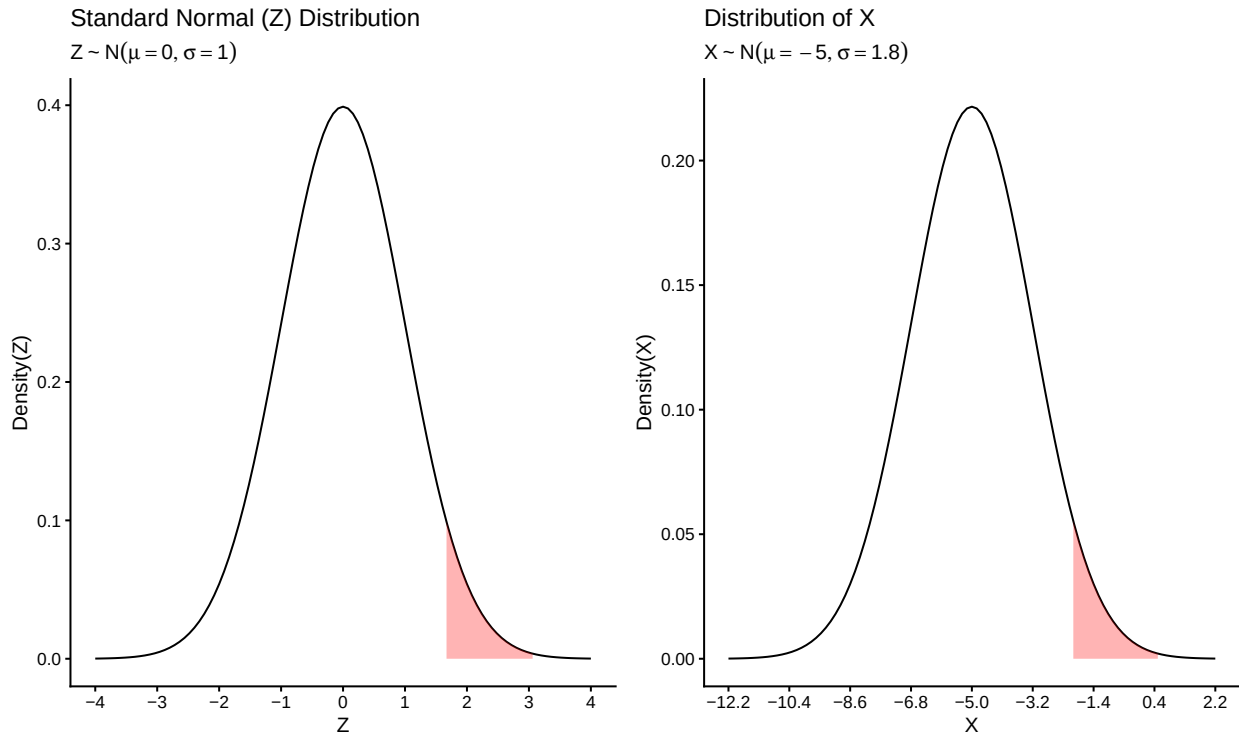
start by finding the z -scores

$$P(-2 < X < 0.5) = P(X < 0.5) - P(X < -2)$$

$$z_{\text{lower}} = \frac{-2 - (-5)}{1.8} = 1.67$$

$$z_{\text{upper}} = \frac{0.5 - (-5)}{1.8} = 3.06$$

$$P(-2 < X < 0.5) = P(1.67 < Z < 3.06)$$



References